

Performance Testing

Date	25 June 2026
Team ID	-
Project Name	Smart lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Flask Local Server (VS Code)
Type of Testing	Functional Testing, Load Testing
Target Module / API	Loan Prediction API (/predict)
Test Environment	Local System (Windows, VS Code, Python 3.x, Flask)
Test Date	10-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single User Prediction	1	10 sec	Loan prediction is displayed successfully.
2	Multiple Concurrent Users	10	30 sec	All requests are processed without errors.
3	Heavy Load Test	5	20 sec	Validation message is displayed to the user.
4	Stress Test	20	60 sec	Application remains stable and responsive.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.18 sec	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	3.42 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	25 Req/sec	31 Req/sec	Pass	Better than target
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	56%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	63%	Pass	Memory usage within acceptable range

Step 4: Observations & Analysis

- The Smart Lender application successfully handled all loan prediction requests without failures.
- Average prediction response time was less than one second.
- The Flask application remained stable under continuous requests.
- CPU and memory utilization remained below the target threshold.
- The machine learning model generated accurate predictions with minimal processing time.
- No significant performance bottlenecks were observed during testing.

Step 5: Screenshots / Evidence

File Edit Selection View Go Run ...

Smart_Lender_AI_ML

index.html app.py

app.py > ...

```
6
7 # Load model and scaler
8 model = joblib.load('models/model.pkl')
9 scaler = joblib.load('models/scaler.pkl')
10
11
12 @app.route('/')
13 def welcome():
14     return render_template('welcome.html')
15
16
17 @app.route('/home')
18 def home():
19     return render_template('index.html')
20
```

PROBLEMS OUTPUT TERMINAL PORTS POSTMAN CONSOLE

TERMINAL

PS C:\Users\LENOVO T450S\OneDrive\Desktop\Smart_Lender_AI_ML> python app.py

```
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production
WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with watchdog (windowsapi)
* Debugger is active!
* Debugger PIN: 108-062-420
127.0.0.1 - - [11/Jul/2026 19:11:29] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [11/Jul/2026 19:11:29] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [11/Jul/2026 19:11:32] "GET /home HTTP/1.1" 200 -
```

DEBUG CONSOLE

Filter (e.g. text, texclude, \...

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Python Python 3.12 (64-bit) Go Live



